

Krishnendu Karmakar

Senior Backend Engineer | Golang | Cloud & Distributed Systems
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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years of experience in Golang and distributed systems, building fault-tolerant, scalable SMS delivery platform, products and real-time inventory applications. Skilled in gRPC, REST, GraphQL, SQL/NoSQL, messaging queues, and AWS, with strong knowledge of Kubernetes, Docker, and CI/CD. Uses AI-assisted workflows (Cursor, Claude) to accelerate development and code review.

PROFESSIONAL EXPERIENCE

Kaleyra (CPaaS Software)

Senior Software Engineer · Jun 2022 – Present · Remote, India
Promoted from Software Engineering Professional to Senior Software Engineer

High-Throughput SMS Delivery Platform

- **Build and optimised a global SMS delivery system processing 6,000+ RPS for enterprise clients across 190+ countries sustaining 50M+ requests/day at peak load.**
- Designed and maintained 10+ public REST and internal gRPC APIs, enforcing strict validation, backward compatibility, and comprehensive error handling.
- **Migrated RabbitMQ and Redis workers from PHP to Golang using Go's concurrency models (Worker pool & Fan-out) — cut message processing latency by ~80–90% reaching below 5ms latency.**
- Implemented priority-based message processing by SMS route type (OTP, transactional, marketing), with elevated priority handling for premium customer traffic to ensure faster delivery for time-sensitive messages.

Active-Active Disaster Recovery — Multi-Region AWS Architecture

- Contributed to dual-region AWS deployments across India (ap-south-1/2) and Southeast Asia (ap-southeast-1, ap-east-1), ensuring high availability and geographic redundancy with live traffic across all sites.
- Engineered a custom Redis Cache Moderator — a coordination layer maintaining Redis data consistency across geographically distributed deployments without Redis Enterprise, eliminating vendor dependency.
- Collaborated with database teams to implement bi-directional synchronisation across MySQL and MongoDB clusters ensuring consistency without service interruption.

Cross-Team API Integration

- Led integration of 5 cross-team APIs (Auth, Catalogue, Gateway, Billing, Callback) into the SMS platform for validations, data transitions, and balance deductions — coordinating across multiple internal teams.
- Implemented retry and fallback handling for API failures, ensuring SMS billing and validation pipelines remained operational during upstream outages.

Two-Way SMS Configurable System

- **Coded and contributed to the architecture of a configurable two-way messaging system supporting automated replies and dynamic workflows, with zero-downtime deployments operating at max 1000 RPS across 190+ countries processing 3 billion request monthly with average latency of 1.5–3 seconds.**
- Developed 5 internal CRUD REST APIs and 5 RabbitMQ workers orchestrating emails, SMS, webhooks, and subscriptions based on configurable customer keyword.

Multi-Channel OTP Verification System

- **Built a fault-tolerant OTP service across SMS, Email, and Voice Call with retry and timeout mechanisms at every delivery layer processing 10+ million requests daily.**
- Replaced Redis with DynamoDB as primary state store for OTP verification, achieving eventual consistency and eliminating cache-layer failures under cross-region load.
- Integrated AWS SQS as a buffering layer between OTP generation and delivery services, improving throughput and isolating failure domains.

Engineering Practices & Platform Reliability

- Established structured logging (Uber Zap) and integrated New Relic for APM and event monitoring, alongside real-time request tracking — reducing average debugging time by ~60%.
- Implemented AES-256 encryption/decryption and data masking for PII, ensuring data-protection compliance.
- Built an API-key-based authorization middleware for request-level access control.

Computer Assembly Management Platform — HP (Client Project)

Inventory & Component Selection Application

- Built a platform for HP teams to view real-time component inventory (CPUs, monitors, keyboards etc) across factories all over the world, supporting component selection and PDF-based processing requests routed via email.
- Developed 3 GraphQL, 5 gRPC APIs in Golang for frontend-backend, microservice communication, powering real-time inventory queries and component selection workflows.
- Designed and implemented a RBAC system leveraging a streamlined two-table architecture, enabling strict permission enforcement and secure access management in the application.

Excel-to-PostgreSQL Data Import CLI

- Worked on a Golang CLI tool to validate and import Excel datasets into PostgreSQL, eliminating ~95% of manual data entry effort in component inventory.
- Designed and developed 4 CRUD REST and 8 gRPC APIs to optimize inventory management; achieved 80%+ unit test coverage across 2 repositories, significantly reducing regression issues.

SKILLS & TECHNOLOGIES

Languages	Golang (primary) · Python · Rust (learning) · PHP
APIs & Protocols	REST · gRPC · GraphQL · WebSockets
Databases	PostgreSQL · MySQL · MongoDB · Redis · DynamoDB
Message Brokers	RabbitMQ · AWS SQS · Redis Queue
Cloud & DevOps	AWS (SQS, EC2, S3, IAM, VPC, ELB, DynamoDB) · Azure · Docker · Kubernetes · Ansible · Jenkins · CI/CD · Terraform · Linux
Architecture	Microservices · Event-Driven Design · Rate Limiting · API Gateway · LLD/HLD · Distributed Systems
AI & Dev Tools	Cursor · Claude · GitHub Copilot · AI-assisted development · Insomnia · Postman · Draw.io

KEY PROJECTS (OPEN SOURCE)

Kube-Pulse (Active) | Go · Rust · gRPC · Kubernetes

Building Microservice-based Kubernetes observability system for developers. Services communicate over gRPC.

- Gateway (Go/REST): github.com/krish8learn/kube-pulse-gateway
- Collector (Rust/k8s metrics): github.com/krish8learn/kube-pulse-collector
- Analyser (Rust/anomaly detection): github.com/krish8learn/kube-pulse-analyzer

EDUCATION

Bachelor of Technology — Electronics and Communication Engineering
Kalyani Government Engineering College, MAKAUT University · CGPA: 8.5 · 2017–2020

CERTIFICATIONS

- Executive PG Certification — Cloud Computing and DevOps | IIT Roorkee / Intellipaat · Mar 2026
- Rust Fundamentals | Duke University / Coursera · Apr 2026
- Golang Master Certification | Udemy · Jan 2021
- PostgreSQL Certification | Udemy · Feb 2021
- Python Certification | Coursera / University of Michigan · Sep 2020
- MongoDB Basic Course | MongoDB University · Mar 2022

LANGUAGES

Bengali (Native) · English (Fluent) · Hindi (Fluent) · German (Basic / A2)